

Chapter-07

1. What does it mean for a wireless network to be operating in “infrastructure mode”? If the network is not in infrastructure mode, what mode of operation is it in, and what is the difference between that mode of operation and infrastructure mode?

Infrastructure mode is when a wireless network has a central device, like a base station or an access point (AP), that all the wireless devices connect to. This central device acts as a bridge, connecting the wireless devices to a larger network, often a wired one like the internet. In this mode, wireless devices don't talk to each other directly; they all communicate through the **Access Point**. A good example of this is your home Wi-Fi network where all your devices (phone, laptop, smart TV) connect to your router, which is the AP.

If a wireless network is **not** in infrastructure mode, it's operating in **ad-hoc mode**. In ad-hoc mode, there is no central access point. Instead, the wireless devices connect directly to each other to form a temporary network. Think of it like a small group of laptops connecting wirelessly to share files without needing a Wi-Fi router.

The key difference is that infrastructure mode provides centralized coordination and usually better performance and reach, while ad hoc mode is decentralized and suitable for small, short-term peer-to-peer connections.

2. In 4G/5G networks, what effect will handoff have on end-to-end delays of datagrams between the source and destination?

Handoff is the process of a mobile device switching from one base station to another as it moves. This action can cause a temporary increase in the end-to-end delay of datagrams. When a handoff occurs, the network needs time to reroute the data path from the old base station to the new one, and during this brief period, datagrams might be lost or temporarily held up. The end-to-end delay can also increase because the time it takes to set up the new connection adds a bit of latency, and the new data path might not be as optimal as the old one.

However, modern 4G and 5G networks are designed to minimize this disruption. The handoff process is made very fast and seamless to limit the impact on the user's connection. Despite the temporary delay, these networks ensure that the user's service remains uninterrupted.

3. Suppose the IEEE 802.11 RTS and CTS frames were as long as the standard DATA and ACK frames. Would there be any advantage to using the CTS and RTS frames? Why or why not?

The main purpose of the RTS (Request to Send) and CTS (Clear to Send) frames is to solve the hidden terminal problem, especially in situations where long data frames are being transmitted. They do this by reserving the channel for the sender and receiver for a short time, preventing other devices from interfering.

If the RTS and CTS frames were as long as the standard DATA and ACK frames, then they would not be useful. The main idea of RTS/CTS is that they are much shorter than a normal data frame. By sending these short control frames first, a device can quickly check and reserve the channel with very little cost. If the RTS/CTS frames were long, then a collision between two devices' RTS/CTS frames would waste just as much time as a collision between full data frames. In other words, we would only be replacing one long collision with another, which removes the whole benefit of using short control frames in the first place.

4. What is the role of the base station in 4G/5G cellular architecture? With which other 4G/5G network elements (mobile device, MME, HSS, Serving Gateway Router, PDN Gateway Router) does it directly communicate with in the control plane? In the data plane?

The **base station** (also known as a eNodeB in 4G and a gNodeB in 5G) is the radio tower that the mobile device communicates with. Its primary role is to manage the radio link between the mobile device and the rest of the cellular network. It handles the physical transmission and reception of signals and is responsible for managing things like handovers.

- **In the control plane:** The base station directly communicates with the **MME (Mobility Management Entity)**. The MME is the central brain of the control plane, handling things like registration, authentication, and mobility management (like handovers).
- **In the data plane:** The base station directly communicates with the **Serving Gateway Router**. The Serving Gateway is responsible for routing user data (datagrams) between the base station and the PDN Gateway, which connects to the internet. The base station's job is to forward the data it receives from the mobile device to the Serving Gateway.

5. Does the LTE wireless access network use FDMA, TDMA, or both? Explain your answer.

The LTE wireless access network uses a combination of both FDMA (Frequency Division Multiple Access) and TDMA (Time Division Multiple Access) in a more advanced form called **OFDMA (Orthogonal Frequency Division Multiple Access)** and **SC-FDMA (Single-Carrier Frequency Division Multiple Access)**.

- **FDMA:** This concept is used by dividing the available frequency spectrum into multiple sub-carriers.
- **TDMA:** This is used by allocating these sub-carriers to different users for specific time slots.

OFDMA is used for the downlink (base station to mobile device) and allows the base station to send data to multiple users simultaneously on different sub-carriers. SC-FDMA is used for the uplink (mobile device to base station) and is a variation that helps reduce power consumption for mobile devices. So, while it's not a simple FDMA or TDMA, the underlying principles of both are combined and enhanced in LTE to provide efficient and flexible use of the radio spectrum.

6. a) What is meant by a “visited network” and a “home network” in 4G/5G cellular architecture? b) What is meant by “hand over” of a network device?

a) In 4G/5G cellular architecture:

- **Home network:** This is the network that a mobile device's subscription belongs to. It is responsible for managing the user's account, handling billing, and storing all permanent information about the device and user. This information is kept in a database called the **HSS (Home Subscriber Server)**.
- **Visited network:** This is a different network that a mobile device connects to when it is roaming outside its home network's coverage area. The visited network provides the mobile device with cellular service, but it needs to communicate with the home network to verify the user's subscription and other details.

b) **Handover** (or handoff) is the process of a mobile device seamlessly switching its connection from one base station to another as it moves. This is crucial for maintaining a continuous connection. For

example, when we are on a phone call and driving, our phone will move from the coverage area of one base station to the next, and the network performs a handover so your call doesn't drop.

7. When a mobile device is handed over from one base station to another in a 4G/5G network, which network element makes the decision to initiate that handover? Which network element chooses the target base station to which the mobile device will be handed over?

In a 4G/5G network, the **mobile device** itself plays a key role in initiating the handover process. The device continuously measures the signal strength of its current base station and neighbouring base stations. When the signal from a neighbouring base station becomes stronger than the current one, the mobile device sends a report to its current base station.

Based on this report, the **MME (Mobility Management Entity)** makes the final decision to initiate the handover. The **MME** also coordinates with the different base stations and chooses the target base station. The MME uses information about the network's load, available resources, and the mobile device's measurements to pick the best new base station for the handover.

8. Consider the single-sender CDMA example in Figure 7.5. What would be the sender's output (for the 2 data bits shown) if the sender's CDMA code were (1, 1, -1, 1, -1, -1, 1)?

The sender's output is found by multiplying each data bit by the entire CDMA code.

The data bits are 1 and -1. The CDMA code is (1, 1, -1, 1, -1, -1, 1).

1. For the first data bit, which is 1, the output is:

$$1 * (1, 1, -1, 1, -1, -1, 1) = (1, 1, -1, 1, -1, -1, 1)$$

2. For the second data bit, which is -1, the output is:

$$-1 * (1, 1, -1, 1, -1, -1, 1) = (-1, -1, 1, -1, 1, 1, -1)$$

So, the sender's combined output for the two data bits would be the concatenation of these two sequences:

(1, 1, -1, 1, -1, -1, 1, -1, -1, 1, 1, 1, -1)

Chapter-08

1. Suppose you want to encrypt the message 10010111 by encrypting the decimal number that corresponds to the message. What is the decimal number?

To find the decimal number for the binary message 10010111, we use the following calculation:

$$1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0.$$

So,

$$128 + 0 + 0 + 16 + 0 + 4 + 2 + 1 = 151.$$

The decimal number is **151**.

2. In what way does a hash provide a better message integrity check than a checksum (such as the Internet checksum)?

A hash function provides a much stronger message integrity check than a simple checksum primarily because it is **cryptographically secure**. A checksum, like the Internet checksum, is designed to detect random, accidental errors that occur during transmission. It is relatively easy for a malicious attacker to deliberately alter a message and calculate a new checksum that matches the altered message, thereby fooling the receiver.

A cryptographic hash function, on the other hand, is designed to be one-way and collision-resistant. This means:

1. It is computationally infeasible to find a message that produces a given hash value.
2. It is computationally infeasible to find two different messages that produce the same hash value.

Therefore, if a message is tampered with, even a single bit, the resulting hash will change drastically, making it virtually impossible for an attacker to create a modified message that has the same hash as the original. This makes a hash an effective tool against both accidental and deliberate modifications.

3. What does it mean for a signed document to be verifiable and nonforgeable?

- **Verifiable:** A signed document is verifiable if anyone can use the signer's public key to confirm that the document was indeed signed by the person who claims to have signed it. The verification process proves the authenticity of the signature and the integrity of the document, ensuring it has not been altered since it was signed.
- **Nonforgeable:** A signed document is called nonforgeable if it is practically impossible for anyone except the real signer to create a valid signature for that document. This is guaranteed because the signature is made using the signer's **private key**, which only the signer knows. Without this private key, no attacker can generate a signature that will correctly match when checked with the signer's public key. This means no one else can fake the signer's identity or produce a false signature.

4. Suppose Alice has a message that she is ready to send to anyone who asks. Thousands of people want to obtain Alice's message, but each wants to be sure of the integrity of the message. In this context, do you think a MAC-based or a digital-signature-based integrity scheme is more suitable? Why?¹

A **digital-signature-based integrity scheme** is more suitable in this scenario.

- **Digital signatures** use a public key/private key pair. Alice signs the message with her private key, and anyone can use her publicly known public key to verify the signature. This is ideal when the message needs to be distributed widely to many people who don't have a pre-shared secret key with Alice.
- A **MAC (Message Authentication Code)**, on the other hand, relies on a shared secret key between the sender and the receiver. For a MAC-based scheme to work, Alice would need to share a unique secret key with each of the thousands of people who want the message. This is not practical and becomes a significant key management problem. Furthermore, a MAC does not provide non-repudiation; since both the sender and receiver have the secret key, either could have generated the MAC, so the receiver cannot prove to a third party that the sender was the one who sent the message.

In conclusion, digital signatures are perfect for a broadcast or multi-recipient scenario where many people need to verify the integrity of a message without a pre-shared secret, which is exactly what Alice's situation describes.

5. What is the de facto e-mail encryption scheme? What does it use for authentication and message integrity?

The de facto e-mail encryption scheme is **PGP (Pretty Good Privacy)** and its IETF standard equivalent, **S/MIME (Secure/Multipurpose Internet Mail Extensions)**. Both are widely used for securing email.

For **authentication and message integrity**, PGP and S/MIME use a **digital signature**.

- The sender first computes a cryptographic hash of the message.
- This hash is then encrypted with the sender's private key, which serves as the digital signature.
- The receiver uses the sender's public key to decrypt the hash, and then computes the hash of the received message.
- If the two hash values match, it proves that the message has not been altered (integrity) and that it was sent by the claimed sender (authentication), since only the sender's private key could have created the valid signature.

6. Suppose Bob initiates a TCP connection to Trudy who is pretending to be Alice. During the handshake, Trudy sends Bob Alice's certificate. In what step of the TLS handshake algorithm will Bob discover that he is not communicating with Alice?

Bob will discover he is not communicating with Alice in the **Client's verification of the server's certificate**.

Here is a more detailed breakdown of the TLS handshake steps:

1. **ClientHello:** Bob sends a ClientHello message to Trudy.

2. **ServerHello:** Trudy, pretending to be Alice, responds with a ServerHello and sends Alice's certificate.
3. **Certificate Validation:** This is the crucial step. Bob receives the certificate and performs a series of checks. He verifies the certificate's validity, checks if it's signed by a trusted Certificate Authority (CA), and importantly, he checks the name on the certificate. Trudy's IP address and domain name will not match the domain name listed in Alice's certificate. For example, if Bob is trying to connect to alice.com, but the IP address he connected to belongs to trudy.net, the certificate for alice.com will not be valid for the connection. This mismatch will be detected by Bob, who will then terminate the connection, realizing that he is not communicating with the legitimate server.

7. Consider RSA with $p = 7$ and $q = 13$.

a. What are n and z ?

b. Let e be 17. Why is this an acceptable choice for e ?

c. Find d such that $de = 1 \pmod{z}$.

d. Encrypt the message $m = 9$ using the key (n, e) . Let c denote the corresponding ciphertext. Show all work.

DRSA Cryptosystem Problem Solution

Problem Statement

Given Parameters:

- Prime numbers: $p = 7, q = 13$
- Public exponent: $e = 17$
- Message: $m = 9$

Part (a): Compute n and z

Computing n (Public Modulus)

$$n = p \times q = 7 \times 13 = 91$$

Computing z (Euler's Totient Function)

$$z = \phi(n) = (p - 1)(q - 1) = (7 - 1)(13 - 1) = 6 \times 12 = 72$$

Answer: $n = 91, z = 72$

Part (b): Validation of $e = 17$

For e to be a valid public exponent, it must satisfy:

Condition 1: Range Check

$$1 < e < z$$

$$1 < 17 < 72 \checkmark$$

Condition 2: Coprimality Check

$$\gcd(e, z) = \gcd(17, 72) = 1$$

Euclidean Algorithm Verification:

$$72 = 4 \times 17 + 4$$

$$17 = 4 \times 4 + 1$$

$$4 = 4 \times 1 + 0$$

Therefore, $\gcd(17, 72) = 1 \checkmark$

Conclusion: $e = 17$ is acceptable because it satisfies both required conditions.

Part (c): Find Private Key d

We need to find d such that: $de \equiv 1 \pmod{z}$

This is equivalent to solving: $17d \equiv 1 \pmod{72}$

Extended Euclidean Algorithm

Forward Pass (Euclidean Algorithm):

$$72 = 4 \times 17 + 4$$

$$17 = 4 \times 4 + 1$$

$$4 = 4 \times 1 + 0$$

Backward Substitution:

$$1 = 17 - 4 \times 4$$

From the first equation: $4 = 72 - 4 \times 17$

Substituting:

$$1 = 17 - 4 \times (72 - 4 \times 17)$$

$$1 = 17 - 4 \times 72 + 16 \times 17$$

$$1 = 17 \times 17 - 4 \times 72$$

Therefore: $17 \times 17 \equiv 1 \pmod{72}$

Private Key: $d = 17$

Verification

$$17 \times 17 = 289$$

$$289 \div 72 = 4 \text{ remainder } 1$$

$$289 \equiv 1 \pmod{72} \checkmark$$

Part (d): Encrypt Message $m = 9$

Encryption Formula: $c \equiv m^e \pmod{n}$

We need to compute: $9^{17} \bmod 91$

Method: Repeated Squaring

Building Powers of 9:

$$9^1 \equiv 9 \pmod{91}$$

$$9^2 \equiv 81 \pmod{91}$$

$$9^4 \equiv (9^2)^2 = 81^2 = 6561 \equiv 9 \pmod{91}$$

$$9^8 \equiv (9^4)^2 = 9^2 = 81 \pmod{91}$$

$$9^{16} \equiv (9^8)^2 = 81^2 = 6561 \equiv 9 \pmod{91}$$

Binary Representation: $17 = 16 + 1$

Final Calculation:

$$9^{17} = 9^{16} \times 9^1 \equiv 9 \times 9 = 81 \pmod{91}$$

Ciphertext: $c = 81$

8. a. Using RSA, choose $p = 5$ and $q = 7$, and encode the numbers 12, 19, and 27 separately. Apply the decryption algorithm to the encrypted version to recover the original plaintext message.

b. Choose p and q of your own and encrypt 1834 as one message m

Part A: RSA with $p = 5$, $q = 7$

Step 1: Calculate RSA Parameters

- $p = 5, q = 7$
- $n = p \times q = 5 \times 7 = 35$
- $z = \phi(n) = (p-1)(q-1) = 4 \times 6 = 24$

Step 2: Choose Public Exponent e

We need e such that $\gcd(e, \phi(n)) = 1$ and $1 < e < \phi(n)$. Let's choose $e = 5$ (since $\gcd(5, 24) = 1$).

Step 3: Calculate Private Exponent d

We need d such that $e \times d \equiv 1 \pmod{z}$

$$= 5 \times d \equiv 1 \pmod{24}$$

Using the extended Euclidean algorithm:

- $24 = 4 \times 5 + 4$
- $5 = 1 \times 4 + 1$
- $4 = 4 \times 1 + 0$

Working backwards:

- $1 = 5 - 1 \times 4$
- $1 = 5 - 1 \times (24 - 4 \times 5)$
- $1 = 5 \times 5 - 1 \times 24$

Therefore, **d = 5** (since $5 \times 5 = 25 \equiv 1 \pmod{24}$)

Step 4: RSA Key Pair

- **Public Key: (n, e) = (35, 5)**
- **Private Key: (n, d) = (35, 5)**

Step 5: Encrypt Messages

Encrypting m = 12:

$$c = m^e \pmod{n}$$

$$c_1 = 12^5 \pmod{35}$$

- $12^2 = 144 \equiv 4 \pmod{35}$
- $12^4 = 4^2 = 16 \pmod{35}$
- $12^5 = 12 \times 16 = 192 \equiv 17 \pmod{35}$ **$c_1 = 17$**

Encrypting m = 19:

$$c_2 = 19^5 \pmod{35}$$

- $19^2 = 361 \equiv 11 \pmod{35}$
- $19^4 = 11^2 = 121 \equiv 16 \pmod{35}$
- $19^5 = 19 \times 16 = 304 \equiv 24 \pmod{35}$ **$c_2 = 24$**

Encrypting m = 27:

$$c_3 = 27^5 \pmod{35}$$

- $27^2 = 729 \equiv 29 \pmod{35}$
- $27^4 = 29^2 = 841 \equiv 1 \pmod{35}$

- $27^5 = 27 \times 1 = 27 \pmod{35}$ **$c_3 = 27$**

Step 6: Decrypt Messages (Verification)

$$m = c^e \pmod{n}$$

Decrypting $c_1 = 17$:

$$m_1 = 17^5 \pmod{35}$$

- $17^2 = 289 \equiv 9 \pmod{35}$
- $17^4 = 9^2 = 81 \equiv 11 \pmod{35}$
- $17^5 = 17 \times 11 = 187 \equiv 12 \pmod{35}$ **$m_1 = 12$ ✓**

Decrypting $c_2 = 24$:

$$m_2 = 24^5 \pmod{35}$$

- $24^2 = 576 \equiv 16 \pmod{35}$
- $24^4 = 16^2 = 256 \equiv 11 \pmod{35}$
- $24^5 = 24 \times 11 = 264 \equiv 19 \pmod{35}$ **$m_2 = 19$ ✓**

Decrypting $c_3 = 27$:

$$m_3 = 27^5 \pmod{35} = 27 \text{ (as calculated above)} \text{ **$m_3 = 27$ ✓**}$$

Summary for Part A:

- Original messages: 12, 19, 27
- Encrypted messages: 17, 24, 27
- Decrypted messages: 12, 19, 27 (verified)

RSA Encryption Example

1. Select Prime Numbers:

- $p = 7$
- $q = 11$

2. Calculate Modulus and Euler's Totient:

- $n = p \times q = 7 \times 11 = 77$
- $\varphi(n) = (p-1) \times (q-1) = 6 \times 10 = 60$

3. Choose Public Key Exponent (e):

- Must satisfy: $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$
- Let's choose $e = 7$ (since $\gcd(7, 60) = 1$)

4. Message Preparation:

- Original message: "1834" (treated as numerical value 1834)

5. Encryption Process:

- Encryption formula: $C \equiv M^e \pmod{n}$
- Calculation: $1834^7 \pmod{77}$

Step-by-Step Modular Exponentiation:

a) First reduce 1834 modulo 77:

- $1834 \div 77 \approx 23.818$
- $77 \times 23 = 1771$
- $1834 - 1771 = 63$
- So, $1834 \equiv 63 \pmod{77}$

b) Now calculate $63^7 \pmod{77}$:

- $63^2 = 3969 \equiv 3969 - 51 \times 77 = 3969 - 3927 = 42 \pmod{77}$
- $63^4 = (63^2)^2 \equiv 42^2 = 1764 \equiv 1764 - 22 \times 77 = 1764 - 1694 = 70 \pmod{77}$
- $63^7 = 63^4 \times 63^2 \times 63^1 \equiv 70 \times 42 \times 63 \pmod{77}$
- $70 \times 42 = 2940 \equiv 2940 - 38 \times 77 = 2940 - 2926 = 14 \pmod{77}$
- $14 \times 63 = 882 \equiv 882 - 11 \times 77 = 882 - 847 = 35 \pmod{77}$

c) Final result: $1834^7 \pmod{77} \equiv 35 \pmod{77}$

6. Encrypted Message:

- The ciphertext is 35

